

Customer Shopping Behavior Analysis

End-to-End Data Analytics Project — Python (EDA) • SQL Server (Business Querying) • Power BI (Dashboarding)

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Executive Summary

This project analyzes 3,900 customer transactions from a retail apparel dataset to understand purchasing behavior across demographics, product categories, and customer loyalty tiers. The workflow spans the full analytics stack expected of a data analyst role: data cleaning and feature engineering in Python, a relational data pipeline into SQL Server, ten business-driven SQL queries, and an interactive Power BI dashboard for stakeholder consumption.

The analysis generated \$233,081 in total tracked revenue across 3,900 customers, an average order value of \$59.76, and an average product review rating of 3.75 / 5. The clearest business signal in the data is customer loyalty concentration: 70% of customers qualify as “Loyal” (16+ prior purchases), but only 27% are active subscribers — pointing to a meaningful gap between purchase loyalty and subscription conversion that a retention campaign could target directly.

3,900	\$233,081	\$59.76	3.75 / 5
Customers Analyzed	Total Revenue	Avg. Order Value	Avg. Review Rating

Project Overview

Objective: Identify the customer, product, and behavioral factors driving revenue and loyalty in a retail apparel business, and package the findings into a decision-ready dashboard.

Dataset: customer_shopping_behavior.csv — 3,900 rows × 18 columns covering customer demographics (age, gender, location), transaction details (item, category, purchase amount, season), engagement signals (review rating, subscription status, previous purchases, purchase frequency), and fulfillment details (shipping type, discount/promo usage, payment method).

Tools & workflow:

- **Python (pandas)** — data cleaning, missing-value treatment, feature engineering, and loading the cleaned dataset into SQL Server via SQLAlchemy
- **Microsoft SQL Server** — ten business questions answered with GROUP BY aggregations, CTEs, window functions (RANK/ROW_NUMBER), and subqueries
- **Power BI** — a single-page interactive dashboard with KPI cards, category/season/age breakdowns, and cross-filtering slicers

Data Preparation & Methodology

Before analysis, the raw dataset was profiled and cleaned in Python:

- **Missing values:** Null review ratings were imputed using the median rating within each product category, preserving category-level rating distribution rather than skewing toward a single global average.
- **Column standardization:** Headers were normalized to snake_case (e.g. “Purchase Amount (USD)” → purchase_amount) for consistency across the Python, SQL, and Power BI layers.
- **Feature engineering — age_group:** Customers were binned into four equal-sized quartile groups (Young Adults 18–31, Adults 32–44, Middle-Aged 45–57, Seniors 58–70) to enable cohort-level comparison.
- **Feature engineering — frequency_days:** The categorical “Frequency of Purchases” field (e.g. “Weekly”, “Quarterly”) was mapped to a numeric day-interval, making purchase cadence usable in quantitative analysis.
- **Redundancy check:** discount_applied and promo_code_used were found to match on 100% of records, confirming the two fields encode the same event. The redundant promo_code_used column was dropped before loading to SQL Server.
- **Data pipeline:** The cleaned DataFrame was pushed directly into a SQL Server database (customer_behaviour) via SQLAlchemy, replacing the need for manual CSV re-import and keeping the SQL layer in sync with the cleaned data.

Key Findings

Ten business questions were answered directly in SQL against the cleaned dataset. Results below are reproduced from those queries.

1. Revenue by Gender

Male customers generated \$157,890 in revenue versus \$75,191 from female customers. This gap tracks the customer base composition (2,652 male vs. 1,248 female — a 68/32 split) rather than a difference in spending behavior: average order value is nearly identical across genders (\$59.54 male vs. \$60.25 female). **Implication:** revenue strategy should focus on acquisition mix, not gender-targeted pricing.

2. Revenue by Product Category

Clothing is the dominant revenue driver at \$104,264 (45% of total revenue), followed by Accessories (\$74,200), Footwear (\$36,093), and Outerwear (\$18,524). Outerwear's low share may reflect seasonal demand concentration rather than weak product-market fit.

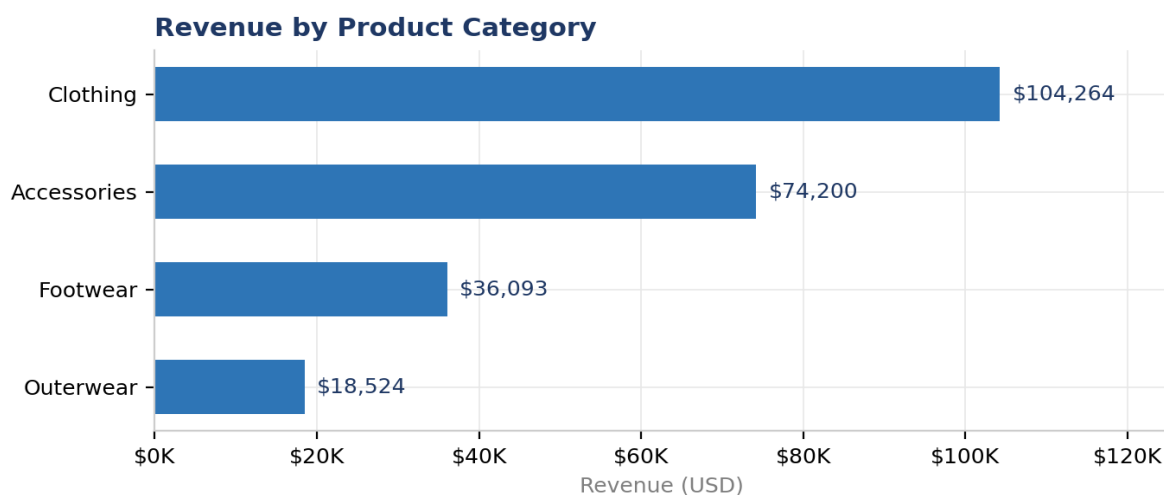


Fig. 1 — Total revenue by product category (n = 3,900 transactions)

3. Top-Rated Products

By average customer review rating, the top five products are:

Rank	Product	Avg. Review Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Handbag	3.78

4. Shipping Type and Order Value

Customers using Express shipping spend slightly more per order (\$60.48 avg.) than those using Standard shipping (\$58.46 avg.) — a \$2.02 gap. The difference is directional but modest, suggesting shipping preference is not a strong lever for increasing basket size on its own.

5. Subscriber vs. Non-Subscriber Spend

Segment	Customers	Avg. Purchase	Total Revenue
Non-Subscribers	2,847	\$59.87	\$170,436
Subscribers	1,053	\$59.49	\$62,645

Subscribers do not spend more per order than non-subscribers (\$59.49 vs. \$59.87). Subscription status appears to function as a retention/loyalty program rather than a spend-driving mechanism — its value case should be framed around retention economics, not incremental order value.

6. Products Most Reliant on Discounts

Hats (50.0%), Sneakers (49.7%), Coats (49.1%), Sweaters (48.2%), and Pants (47.4%) have the highest share of purchases made with a discount applied — roughly half of all units sold in these lines moved at a discount, which is worth monitoring for margin impact.

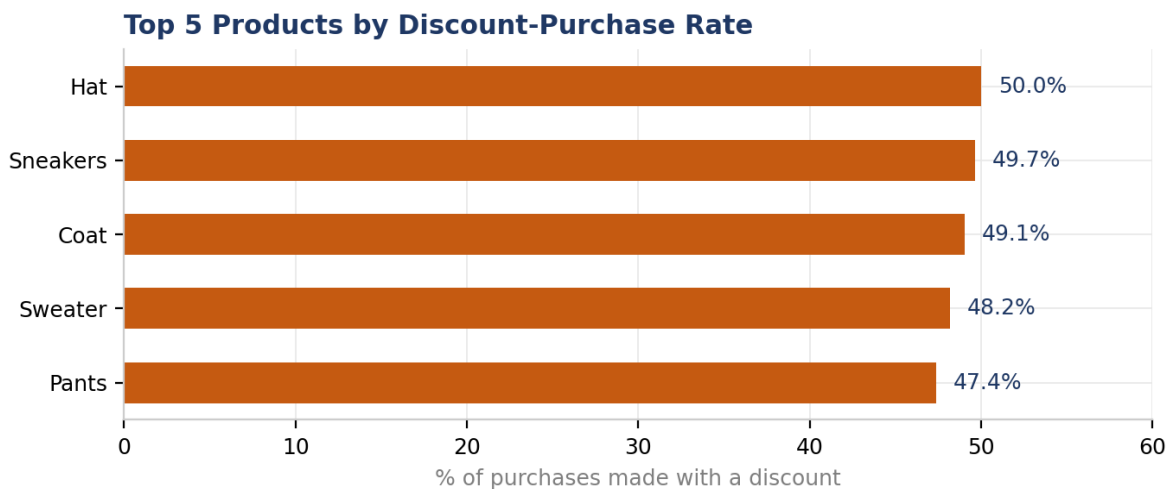


Fig. 2 — Share of purchases made with a discount, top 5 products

7. Customer Segmentation (New / Returning / Loyal)

Segmenting customers by number of previous purchases (New: 1–3, Returning: 4–15, Loyal: 16+) shows the base is heavily weighted toward loyal, repeat customers: 2,721 Loyal (70%), 933 Returning (24%), and only 246 New (6%). A thin New-customer segment relative to a large Loyal base suggests the business is running more on retention than fresh acquisition — worth validating against marketing spend data.

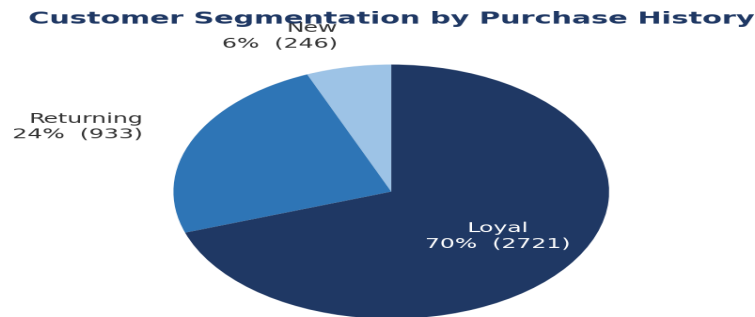


Fig. 3 — Customer distribution across loyalty segments

8. Best-Selling Products by Category

Using a window-function ranking (ROW_NUMBER partitioned by category), the top 3 products per category by order volume are:

Category	#1	#2	#3
Accessories	Jewelry (171)	Belt (161)	Sunglasses (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	—

9. Do Repeat Buyers Subscribe?

Among customers with more than 5 previous purchases, 2,518 (72%) are **not** subscribed, versus 958 (28%) who are. Purchase frequency alone is a weak predictor of subscription — a large pool of proven repeat buyers remains unconverted to the subscription program, representing the clearest actionable opportunity in this dataset.

10. Revenue by Age Group

Revenue is broadly even across age cohorts, with Young Adults (18–31) contributing the most at \$62,143, followed by Middle-Aged (\$59,197), Adults (\$55,978), and Seniors (\$55,763). The near-parity across groups (within an 11% band) indicates the product line has broad-based, age-agnostic appeal rather than a single core demographic.

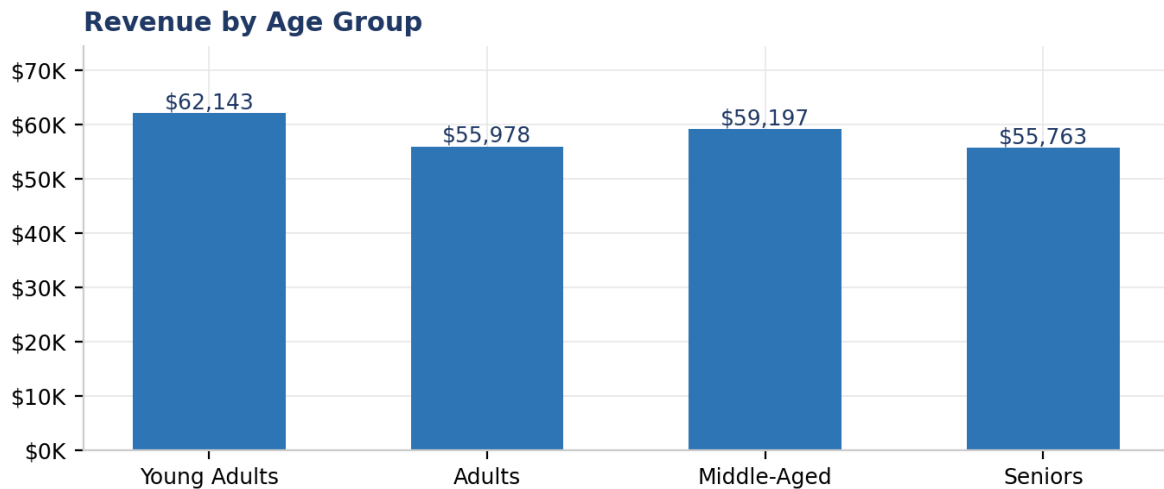


Fig. 4 — Total revenue by age cohort

Power BI Dashboard

The SQL-layer findings were consolidated into a single-page interactive Power BI report (“Customer Behavior Dashboard”) so a non-technical stakeholder can explore the same data without writing queries.

Dashboard components:

- **KPI cards** — Number of Customers, Average Purchase Amount, and Average Review Rating as headline metrics
- **Clustered column chart** — revenue by product category
- **Clustered bar charts** — revenue by age group, and customer count by age group
- **Pie chart** — revenue by season
- **Donut chart** — customer count by subscription status
- **Slicers** — gender, category, subscription status, and shipping type, enabling cross-filtered, self-serve exploration

This mirrors standard BI practice: SQL is used for precise, auditable business-question answers, while Power BI provides the interactive layer for ongoing stakeholder use.

Recommendations

- **Target subscription conversion, not acquisition, first.** 72% of high-frequency repeat buyers (5+ purchases) are unsubscribed. Converting even a fraction of this group is lower-cost than acquiring new customers and directly grows the loyalty program.
- **Audit discount dependency on Hats, Sneakers, and Coats.** With ~48–50% of units moving at a discount, these lines warrant a margin review to confirm the discount rate is intentional and not silently eroding profitability.
- **Re-evaluate the subscription value proposition.** Since subscribers and non-subscribers spend almost identically per order, the subscription program should be positioned and measured on retention/frequency benefits, not average order value.

- **Investigate the New-customer gap.** Only 6% of the customer base falls in the New segment. Cross-referencing this against marketing spend or channel data would clarify whether this reflects a mature customer base or a stalled acquisition funnel.
- **Treat Outerwear as a seasonal, not underperforming, category.** Its lower revenue share likely reflects limited seasonal window rather than weak demand; before reallocating budget away from it, confirm using season-level performance within Outerwear specifically.

Skills Demonstrated

Area	Techniques Applied
Data Cleaning (Python/pandas)	Missing-value imputation by group, column standardization, feature engineering (binning, mapping), redundancy detection
Database & SQL	SQLAlchemy pipeline to SQL Server; GROUP BY aggregation, subqueries, CTEs, window functions (ROW_NUMBER)
Business Analysis	Translating open business questions into quantifiable metrics and actionable recommendations
Data Visualization (Power BI)	KPI cards, multi-chart dashboard design, cross-filtering slicers

Report generated from analysis.ipynb (Python/pandas EDA), Customer_ques.sql (SQL Server business queries), and Customer_Behaviour.pbix (Power BI dashboard). Dataset: customer_shopping_behavior.csv (3,900 rows).